



Taha Rasouli

Machine Learning Engineer

Profile

Machine Learning engineer with end-to-end experience across the ML lifecycle – architecture, training, evaluation, deployment – applied to NLP/LLM systems and autonomous driving/ADAS. Focus: agents that decompose tasks into verifiable steps, and evaluation methods that catch when a model’s stated reasoning isn’t actually grounded.

Education

Apr 2023–2026 **M.Sc. Computer Science (AI-NLP)**,  University of Stuttgart, Germany, Thesis Grade: 1.0
Thesis: Generative AI as the reasoning faculty for autonomous driving Supervisor Dr. Halimeh Agh; examiner Prof. Dr. Jonas Kuhn.

Skills

Tools & Infrastructure	Python, C++, ROS2, SQL, Git, Docker, Azure, Linux; CI/CD; REST API
Fine-Tuning & Deployment	DoRA/LoRA/QLoRA, PEFT, knowledge distillation, latency-constrained inference
Generative Models	CVAE, GAN, diffusion, transformers, Mamba SSMs, GNNs
LLM Agents	LangGraph/LangChain, task decomposition, tool use, prompt engineering
RL & Imitation Learning	continuous-action RL, imitation learning, physical-feasibility-constrained policies
Simulation & Planning	closed-loop simulation (nuScenes/NuPlan), trajectory prediction, multi-agent modeling
Computer Vision	depth/distance estimation, detection & segmentation; Detectron2/MMDetection/YOLO, CNNs, ViTs

Related Projects

- Dec 2025–Jun 2026 **ODIN — LLM Reasoning Layer for Autonomous Driving**, *Master’s Thesis*
Designed two **neural architectures** – **ODIN-Traj**, a Mamba state-space trajectory generator with CVAE stochastic sampling and multi-scale GAN adversarial fine-tuning, and **ODIN-Maneu**, a Bi-GRU maneuver classifier (88.8% accuracy) – plus **ODIN-Plan**, an LLM planning layer built by **fine-tuning Llama-3-8B via DoRA** (0.26% of base params) on DeepSeek-R1-distilled reasoning traces. Closed-loop evaluation on nuScenes (2 Hz) uncovered “reasoning reward-hacking” – 29/30 fluent justifications vs. only 8/30 correctly grounded ($p = 0.0023$, McNemar’s).
- 2024–Present **Imitation-Learning Trajectory Prediction**, *ARENA 2036 / TU Munich*
Multi-agent trajectory/future-prediction model combining **imitation learning** with explicit physical-feasibility constraints; minADE 0.023, minFDE 0.01 vs. baselines; built training/eval pipeline with TU Munich.

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in [taha-rasouli](https://taha-rasouli.github.io)

Professional Experience

- Apr–Sep 2025 **AI ADAS Intern**,  *Porsche Engineering Services GmbH*, Bietigheim-Bissingen
Designed deep learning model architecture for safety-critical scenario prediction; applied **reinforcement learning** for trajectory optimization in a Level 2+ production ADAS pipeline; built validation strategies for real-world deployment reliability.
- 2025–Present **Student Research Assistant, Civil Engineering ML**,  *KIT*, Karlsruhe
Applying ML to civil-engineering optimization problems.
- Jun 2024–Apr 2025 **Student Research Assistant, Agentic AI**,  *Fraunhofer IPA*, Stuttgart
Built LLM/agent prototypes exploring task decomposition, tool use, and multi-step reasoning; deployed scalable MLOps infrastructure (Docker, Azure) for research-scale experimentation.
- 2024–Present **Working Student, ADAS ML**,  *ARENA 2036 / TU Munich*, Stuttgart
Multi-agent trajectory-prediction model via **imitation learning** with physical-feasibility-constrained policy learning (see Related Projects).
- Aug 2023–Present **Student Research Assistant, NLP**,  *ISW, University of Stuttgart*
Designed multi-stage ML pipelines integrating heterogeneous industrial data; applied LLM-based **semantic matching** to harmonize disparate schemas; co-authored 2 IEEE papers.
- 2021–2023 **Machine Learning Engineer**,  *Afarinesh IELTS School*, Tehran
Neural network for auto-grading against human-annotated rubrics; **speech-recognition** model for spoken-language proficiency; LLM-based exercise generation and learner profiling.

Publications

- 2025 S. Ajdinovic, **T. Rasouli**, N. Maisch, A. Wortmann. “AI-Powered Semantic Matching and Data Harmonization for Industrial Applications.” *IEEE Xplore*.
- Forthcoming S. Ajdinovic, **T. Rasouli**, N. Maisch, A. Wortmann. “Semantic Matching and Computation-Enabled Integration of OPC UA Data into Asset Administration Shells using LLMs.”

References

Prof. Dr. Jonas Kuhn (Univ. Stuttgart, IMS) – jonas.kuhn@ims.uni-stuttgart.de
Dr. Halimeh Agh (Univ. Stuttgart, ISTE) – halimeh.agh@iste.uni-stuttgart.de
Ehsan Sharafian (Porsche Engineering) – ehsan.sharafian_ardakani@porsche-engineering.de
Samed Ajdinovic (Univ. Stuttgart, ISW) – samed.ajdinovic@isw.uni-stuttgart.de

Languages

English	C2	<i>IELTS 8.0</i>
German	A2	
Persian	Native	